Discrete random variable probabilit[redacted]tributions

1

a Yes, because $\Sigma P\left(x\right)=1, X$ is discrete numerical, and $0\leq P\left(x\right)\leq 1$.

b No, because even though X is discrete numerical, the other two conditions where $\Sigma P\left(x\right)=1$ and $0\leq P\left(x\right)\leq 1$ are not satisfied.

2

a No

b Yes

c No

d Yes

3

a

i

- $0\leq P\left(x\right)\leq 1$
- $\Sigma P\left(x\right)=1$
- X is discrete numerical.

ii Yes

b

i

- $0\leq P\left(x\right)\leq 1$
- X is discrete numerical.

ii No

c

i

- $0\leq P\left(x\right)\leq 1$
- $\Sigma P\left(x\right)=1$
- X is discrete numerical.

ii Yes

d

i

- $0\leq P\left(x\right)\leq 1$
- X is discrete numerical.

ii No

4

a

x	1	2	3
$P(X=x)$	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$

b Yes, because $\Sigma P\left(x\right)=1, X$ is discrete numerical, and $0\leq P\left(x\right)\leq 1$.

5

a

x	1	2	3
$P(X=x)$	$-\frac{1}{3}$	$-\frac{2}{3}$	$-\frac{3}{3}$

b No, because even though X is discrete numerical, the other two conditions where $\Sigma P(x)=1$ and $0 \leq P(x) \leq 1$ are not satisfied.

6

a

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
$P(X=x)$	0	$\frac{1}{128}$	$\frac{1}{32}$	$\frac{9}{128}$	$\frac{1}{8}$

b No, because even though X is discrete numerical and $0 \leq P(x) \leq 1$, the condition $\Sigma P(x)=1$ is not satisfied.

7

a $\frac{1}{4}$ **b** $\frac{7}{16}$ **c** $\frac{7}{16}$ **d** $\frac{5}{16}$

e $\frac{3}{16}$ **f** $\frac{13}{16}$

8

a

x	0	1	2	3	4
$P(X \leq x)$	0.1	0.3	0.65	0.95	1
$P(X=x)$	0.1	0.2	0.35	0.3	0.05

b

i 0.65 **ii** $\frac{13}{14}$

9

a

x	$P(X \le x)$	$P(X = x)$
0	0.001	0.001
1	0.011	0.010
2	0.055	0.044
3	0.172	0.117
4	0.377	0.205
5	0.623	0.246
6	0.828	0.205
7	0.945	0.117
8	0.989	0.044
9	0.999	0.01
10	1	0.001

b

i 0.172 **ii** 0.055 **iii** 0.066 **iv** 0.912

10

a $c = 60$ **b** $a = 2$ **c** $b = 3$

11

a $k = 0.2$ **b** $k = 0.3$

12

a $k = \frac{1}{11}$ **b** $\frac{4}{11}$ **c** $\frac{8}{11}$ **d** $\frac{4}{9}$

e $\frac{8}{9}$

13

a $k = \frac{20}{31}$ **b** $k = \frac{1}{50}$ **c** $k = 19$ **d** $k = \frac{576}{487}$

14

a

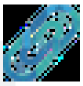
x	0	1	2	3
$P(X = x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

b

x	0	1	2	3
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{5}{12}$

c

x	0	1	2	3
$P(X=x)$	$\frac{1}{13}$	$\frac{2}{13}$	$\frac{2}{13}$	$\frac{8}{13}$



d

x	0	1	2	3
$P(X=x)$	$\frac{1}{14}$	$\frac{5}{14}$	$\frac{5}{14}$	$\frac{3}{14}$

15

a $a=0.23$

b $b=0.16$

16

a $c=0.22$

b $a=0.24$

c $b=0.21$

17

a $k=\frac{1}{15}$

b

x	4	5	6	7	8
$P(X=x)$	$\frac{5}{15}$	$\frac{4}{15}$	$\frac{3}{15}$	$\frac{2}{15}$	$\frac{1}{15}$

c $\frac{4}{5}$

d $\frac{2}{5}$

e $\frac{2}{3}$

f $\frac{9}{14}$

18

a

x	0	1	2	3	4	5
$P(X=x)$	0.01024	0.07680	0.23040	0.34560	0.25920	0.07776

b Negatively skewed

c 0.33696

d 0.98976

e 0.65955

19

a Geometric

b $\frac{1}{2}$

c $\frac{1}{2}$

d 1

e $\frac{7}{32}$

f $\frac{508}{511}$

Uniform distribution

20

a

x	0	2	4	6	8	10
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

b Yes, because $\Sigma P(x) = 1, X$ is discrete numerical, and $0 \leq P(x) \leq 1$.

21

a $k = \frac{1}{6}$

b $P(X=x) = \begin{cases} \frac{1}{6}; x = 1,2,3,4,5,6 \\ 0, \text{ otherwise} \end{cases}$

22

a $k = \frac{1}{5}$

b $\frac{2}{5}$

c $\frac{4}{5}$

d $\frac{3}{4}$

e $\frac{3}{4}$

23

a $\frac{1}{4}$

b $\frac{1}{2}$

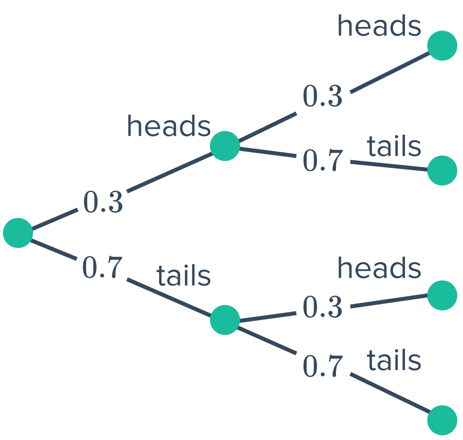
c $\frac{2}{3}$

d 2

Applications

24

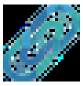
a



b

x	0	1	2
$P(X=x)$	0.09	0.42	0.49

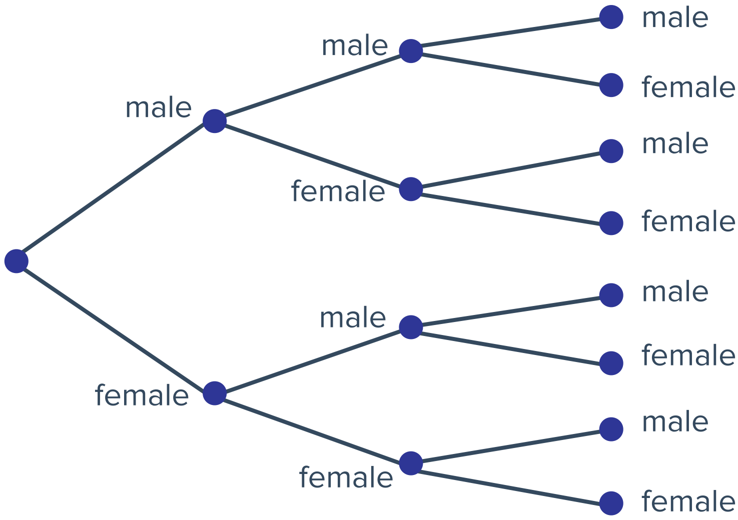
c
- X is discrete numerical
- $0 \leq P(x) \leq 1$
- $\Sigma P(x) = 1$



d Yes

25

a



b

f	0	1	2	3
$P(F=f)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

c Non-uniform

26

a $1, 2, 3, 4, 5, 6$

b Uniform, the outcomes are equally likely.

c

x	1	2	3	4	5	6
$p(x)$	$\frac{996}{6000}$	$\frac{1005}{6000}$	$\frac{994}{6000}$	$\frac{1005}{6000}$	$\frac{996}{6000}$	$\frac{1004}{6000}$

d $\frac{667}{2000}$

e 0.05%

27

a

i $\frac{1}{8}$ ii $\frac{49}{512}$ iii $\frac{343}{4096}$ iv $\frac{117\,649}{2\,097\,152}$

b
- X is discrete numerical.
- $\Sigma P(x) = 1$
- $0 \leq P(x) \leq 1$

c Yes

28

a

x	1	2	3	4	5	6
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

b $\frac{1}{3}$

c $\frac{1}{5}$

29

a $k = \frac{1}{21}$

b

x	1	2	3	4	5	6
$P(X=x)$	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{6}{21}$

30

a 0.1251

b Yes, the distribution is approximately symmetrical with the mean, median and mode coinciding at 10 Jupiter bars.

c

i 0.0948 **ii** 0.0901 **iii** 0.0103 **iv** 0.1198

d Yes the claim is correct as $P(8 \leq X \leq 12) = 0.5713$, which is higher than 50%.

31

a

x	38	39	40	41	42
$P(X=x)$	$\frac{5}{70}$	$\frac{7}{70}$	$\frac{24}{70}$	$\frac{8}{70}$	$\frac{26}{70}$

b

i $\frac{29}{35}$ **ii** $\frac{18}{35}$

33

a $k = 0.8$ **b** 0.8 **c** 0.000256